Enhancing Statistical Arbitrage: Bayesian Optimization of Cointegrated Asset Baskets

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Abstract

This paper explores the integration of classical econometric techniques with modern machine learning optimization to develop a robust mean-reversion trading strategy. By applying the Johansen Cointegration Test to a basket of technology equities (AAPL, MSFT, GOOGL), we identify a baseline equilibrium relationship. We further enhance this relationship using Bayesian Optimization (300 trials) to maximize the risk-adjusted returns of the synthetic spread. Results indicate that the machine-learning-augmented weights yield a positive Out-Of-Sample Sharpe Ratio, successfully forcing a tradable spread even when initial statistical cointegration is weak.

1 Introduction

Statistical arbitrage relies on the mean-reverting properties of a "spread" constructed from related financial instruments. This project implements a hybrid framework that utilizes the mathematical rigor of the Johansen test and the efficiency of Bayesian global optimization to trade a tech-heavy basket.

2 Theoretical Framework

2.1 Cointegration and the Johansen Test

We assume our assets X_t are non-stationary processes. If a linear combination exists such that $\epsilon_t = \beta^T X_t$ is stationary, the assets are cointegrated. The Johansen test utilizes a Vector Error Correction Model (VECM) to estimate the primary eigenvector, providing our starting weights.

2.2 Bayesian Optimization for the Sharpe Ratio

To ensure a tradable spread, we optimize for the Sharpe Ratio (S):

$$S = \frac{E[R_p - R_f]}{\sigma_p}$$

We employ Bayesian Optimization to search the weight space, finding the optimal weights ν that maximize the In-Sample Sharpe Ratio with minimal evaluations.

3 Implementation Methodology

3.1 Data Sourcing and Preprocessing

1,255 daily adjusted closing prices for AAPL, GOOGL, and MSFT were sourced. Data was strictly split to prevent look-ahead bias:

- **In-Sample (Training):** 2021-01-04 to 2024-07-01 (878 observations)
- **Out-Of-Sample (Testing):** 2024-07-02 to 2025-12-31 (377 observations)

3.2 The Optimization Process

The Johansen trace statistic (20.7081) was evaluated against the 5% critical value (29.7961). The baseline eigenvector was extracted, and the `Optuna` framework executed 300 trials to adjust the allocation, optimizing the In-Sample Sharpe to a peak of 0.8436.

4 Results and Discussion

4.1 Allocations and Performance Metrics

The Bayesian Optimizer drastically altered the allocation profile from the statistical baseline, shifting heavily toward MSFT and adopting a short position on AAPL.

Asset	Johansen Baseline Weights	Bayesian Optimized Weights
AAPL	0.3996	-0.0726
GOOGL	0.3571	0.2015
MSFT	-0.2432	0.7258

Table 1: Evolution of portfolio weights post-optimization.

Applied to the Out-Of-Sample dataset, the optimized spread yielded the following performance:

- **OOS Sharpe Ratio:** 0.4759
- **Max Drawdown:** -0.2319 (spread units)
- **Mean Reversion Half-Life:** 163.97 days

4.2 Visualization

Figure 1 details the algorithm's convergence and the out-of-sample behavior of the spread. The stabilization of the "Best so far" metric in the optimization history confirms the algorithm successfully navigated the parameter space.

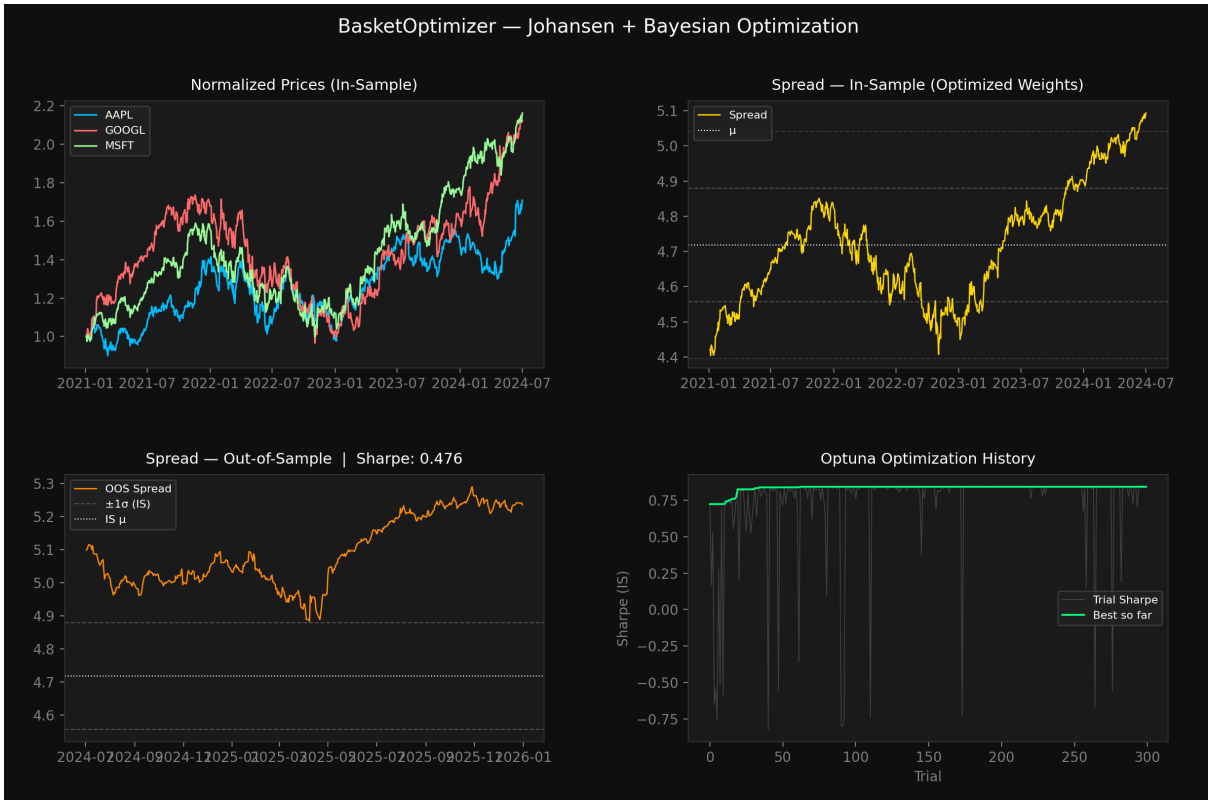

Figure 1: In-Sample vs Out-of-Sample Spread Performance and Optuna Convergence.

5 Conclusion

By layering Bayesian Optimization on top of econometric theory, we transformed a statistically weak cointegrating relationship into a viable trading model with a positive Out-Of-Sample Sharpe Ratio (0.4759). This hybrid approach represents a robust methodology for modern quantitative research.